Instruction Manual - LiDAR data acquisition and simple trajectory implementation

- This manual works to integrate the RPLiDAR C1 to a Ubuntu/Linux device. If your only option is Windows, we encourage you to change the lidar_test.cpp file. It is possible to use a Raspberry Pi and VNCviewer.
- For any question or problem, we encourage visiting and following the instructions from SDK for Slamtec RPLIDAR Github

There are two options to get the LiDAR up and running:

STEP 1

It is easier to just extract directly the rplidar_sdk-master.zip folder that is part of this deliverable and execute a "make" command on the folder "rplidar_sdk-master" and another one on "rplidar_sdk-master/app/lidar_test" . This may however not work due to compatibility problems and different versions.

If it doesn't work, you will have to build the whole environment. To do so, download the sdk from GitHub and execute a "make" command on the main folder. Then, as you go inside of "rplidar_sdk-master", access the folder "app", then create a new folder (we recommend naming it "lidar_test"). Inside of this new folder, make a copy of the provided "lidar_test.cpp" code and name it "main.cpp". Then, include a "makefile" identical to the on "app/ultra_simple" folder. Or else you can just copy/paste the following code:

```
HOME_TREE := ../../

MODULE_NAME := $(notdir $(CURDIR))

include $(HOME_TREE)/mak_def.inc

CXXSRC += main.cpp
C_INCLUDES += -I$(CURDIR)/../../sdk/include -I$(CURDIR)/../../sdk/src

EXTRA_OBJ :=
LD_LIBS += -lstdc++ -lpthread all:

build_app

include $(HOME_TREE)/mak_common.inc clean:

clean_app
```

Then you just have to open a terminal inside the folder you created ("rplidar_sdk-master/app/lidar_test"), and as it has only two files (main.cpp + makefile), you have to run a "make" command.

How to know if it worked? (in both cases)

You have to look into the “rplidar_sdk-master/output/Linux/Release” and see if there is an executable named “lidar_test”. If yes, proceed to step 2.

Please note: lidar_test.exe is designed for the lidar port "/dev/ttyUSB0." Experience shows that this port changes from time to time. Therefore, you must modify the lidar_test's "main.cpp" in app before creating the executable file, "lidar_test.exe."

We've already created several lidar_test.exe files for different ports (lidar_test_USB0.exe, etc.). So, everything above is how we did it (step 1), and now you just need to run these files according to your port (lidar_test_USB*.exe).

STEP 2

Connect the LiDAR using the USB to UART Bridge and connect the USB to the Raspberry or PC. Then, “execute on terminal” the executable “lidar_test” on “rplidar_sdk-master/output/Linux/Release”. It is important to “execute on terminal” once just executing it makes the LiDAR don’t recognize the termination of the program when you unplug the LiDAR from the USB.

STEP 3

This function will allow you to recover data from the LiDAR in a Python code:

```
def get_data():
        measure = []
        start_time = time.time()
        while (time.time() - start_time < 1):
                data=server_socket.recvfrom(1024)
                str_data = data[0].decode("utf-8")[3:].split(" ")
                print(str_data[1]+" "+str_data[3]+'\n')
                angle = float(str_data[1])
                distance = float(str_data[3])
                measure.append((angle,distance))
                return measure
```

This function is crucial for all algorithms using Lidar in the robot.

A better implemented version of this function is available on the “lidar_test.py” which maps the environment.